AFRICAN DEVELOPMENT BANK GROUP
GROUPE DE LA BANQUE AFRICAINE DE DÉVELOPPEMENT

Data Innovation for Africa

NIGHT LIGHTS SERIES · THREE NOTEBOOKS

Night lights: from satellite to statistics

Collect, understand and analyse NASA and NOAA night-time lights data to measure economic activity — one ISO3 code in, a public dashboard out



01 Collect

02 Explore and understand

03 Analyse and publish

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PROGRAMME

Six steps, from the night sky to a published indicator

01
The series
Why lights measure activity, and how the three notebooks fit together

02
Collect
Notebook 1: find, test and download the right NASA Black Marble files

03
Explore and understand
Notebook 2: open the file, decode it, filter it, clip it to the country

04
Analyse
Notebook 3: trends, regions, GDP, shocks, inequality, industrial zones

05
Publish and check
Static dashboard, reproducibility manifest, checks and limits

06
Key takeaways
What to verify before publishing a number

SECTION

01

The series

Why do night-time satellite images tell us about an economy?



Light seen from space, a footprint of activity

~463 m
pixel size (15 arc-seconds)

Every night
since 19 January 2012 (Black Marble)

01:30
local satellite overpass time

What lights capture well

Urban activity, industry running at night, electrification, outages, displacement, sudden shocks

What they capture poorly

Agriculture, daytime trade, much of the services sector and the rural informal economy

How to use them

To complement official statistics: direction, timing and geography of activity — not as a substitute for GDP

Sources: Román et al. (2018); Henderson, Storeygard & Weil (2012); Chen & Nordhaus (2011)

Three notebooks, one pipeline, one parameter

Notebook 1 · Collect

NASA Black Marble (VNP46A1–A4)

- Country footprint and VIIRS tiles
- NASA CMR catalogue query
- Availability matrix, verified download

Output: .h5 files, inventory, log

Notebook 2 · Explore

Anatomy of a Black Marble file

- HDF5 tree, layers, attributes
- Decoding, quality flags, masks
- Mosaic, clipping, distribution

Output: GeoTIFF, CSV, JSON metadata

Notebook 3 · Analyse

VIIRS VNL V2 + Black Marble on Earth Engine

- Trends, regions, convergence
- GDP link, shocks, inequality
- Industrial zones

Output: GitHub Pages dashboard

SECTION

02

Collect

How do we get the right NASA files, and only those?



One ISO3 code in, a verified collection out

1. Environment
2. Dependencies
3. Token (4 sources)
4. Parameters
5. Footprint and tiles
6. CMR queries
7. Availability test
8. Verified download
9. Quality check

Steps 1–7: no account needed

Step 8: Earthdata token

Safe to re-run (idempotent)

NASA Black Marble: four products, one grid

VNP46A1

Daily · raw

At-sensor radiance, uncorrected

Use

Diagnostics

VNP46A2

Daily · corrected

Night corrected for lunar BRDF

Use

Dated events: outages, disasters

VNP46A3

Monthly composite

Clouds and outliers removed

Use

Seasonality, monthly series

VNP46A4

Annual composite

The most stable of the four

Use

Trends, regional comparisons

10° × 10° tile grid | **2,400 × 2,400 pixels per tile** | **nW·cm^{-2}·sr^{-1}** | **HDF-EOS5 (.h5)** | **Collection 2 (v002)**

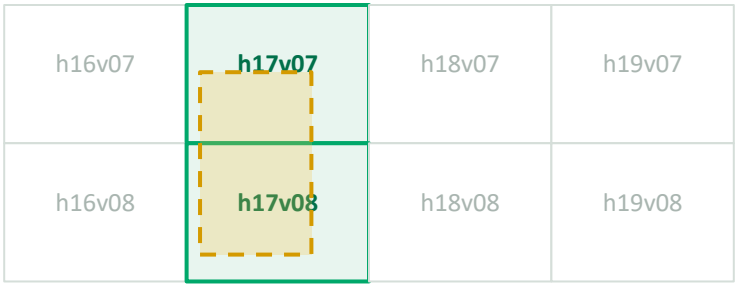
Sources: Román et al. (2018); NASA Black Marble User Guide, Collection 2.0 (2025); LAADS DAAC

From a country code to tiles, with simple arithmetic

Tile index

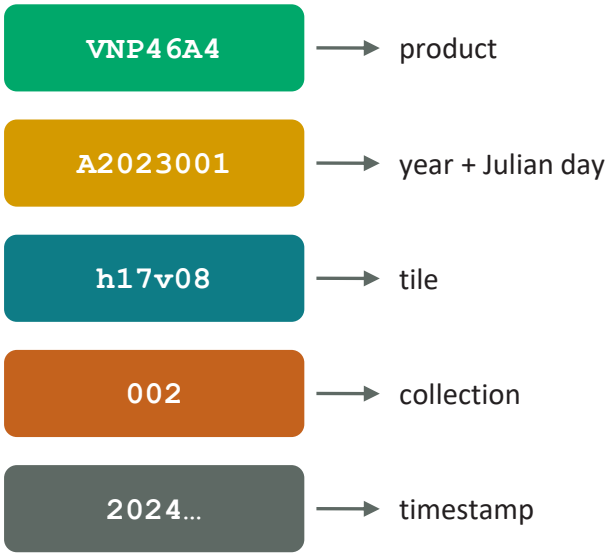
```
h = ⌊(longitude + 180) / 10⌋
v = ⌊(90 − latitude) / 10⌋
```

h: 0 → 35 west to east · v: 0 → 17 north to south

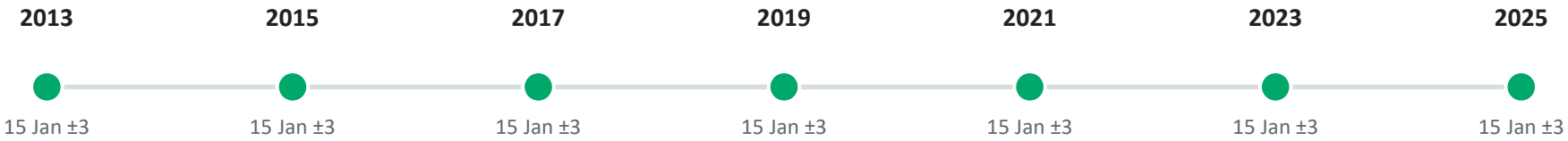


Côte d'Ivoire → h17v07 + h17v08

Anatomy of a file name



Fix the season, vary the year, never a single night



Daily A1 / A2

Fixed day ± 3 nights (7 nights a year), then keep clear, good-quality nights and take their median

Monthly A3

The same month every year: the composite already absorbs clouds and the Moon

Annual A4

The whole year: the right product for year-on-year trends

Moon: 29.5-day cycle — a fixed date does not fix the lunar phase.

Query, measure availability, download

1 · Query

One query per year · pagination (2,000 per page) · de-duplication by tile and date

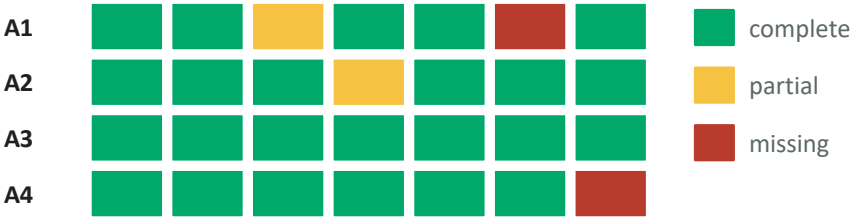
2 · Measure

Expected = windows × tiles × nights · found · completeness · volume in GB

3 · Download

Resume · size check · HDF5 open test · manifest

Coverage matrix (schematic)



Secure by design

The token comes from an environment variable, a .env file, Colab or Kaggle secrets, or a masked prompt — and is only ever sent to *.earthdata.nasa.gov and *.eosdis.nasa.gov.

SECTION

03

Explore and understand

What is really inside a Black Marble file?



A Black Marble file is a workbook, not an image

FILE VNP46A4 … .h5 — *global attributes: sensor, footprint, production date*

HDFEOS / GRIDS / <grid> — *the grid name depends on the product — never hard-code it*

Data Fields — *every useful layer lives here*

Radiance
`AllAngle_Composite_Snow_Free`

Uncertainty
`…_Num` · `…_Std`

Quality
`…_Quality` · `Mandatory_Quality_Flag`

Context
`Land_Water_Mask` · `DNB_Platform`

Coordinates
`lat` · `lon`

Decoding: one equation and one trap

radiance = stored value × scale_factor + add_offset

…after setting every _FillValue pixel to NaN — attributes always read from the file

Daily layer (A2)

`uint16 · × 0.1 · fill 65535`

1247 → 124.7 nW

+6,553.5

per empty pixel if fill is ignored

Composite (A3 / A4, Collection 2)

`float32 · × 1 · fill −999.9`

124.7 → 124.7 nW

−999.9

per empty pixel if fill is ignored

⚠ **Rule:** the fill value is not a zero, let alone a measurement.

Which pixels to trust? Quality flags and masks

Composite quality codes (A3 / A4)

0	Good quality · > 3 nights	kept
1	Poor quality · ≤ 3 nights	dropped
2	Gap-filled from history	dropped
255	Fill value	dropped

QA_MAX = "auto" reads this legend from the file

Land / water mask

Oceans and seas removed: moonlight glint, ships and fisheries are not dark. Without it, coastal and landlocked countries are not comparable.

Cloud mask (daily)

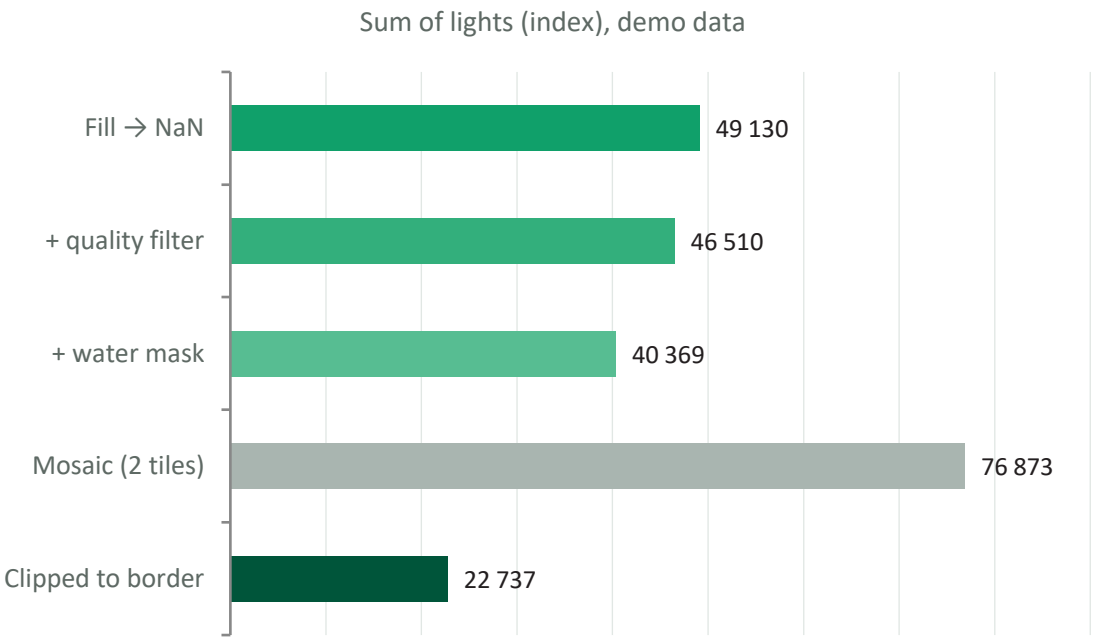
Bits 6–7: confident clear → confident cloudy. Value = (integer >> 6) & 3.

Number of nights

…_Num tells how many nights support each pixel: map it next to the radiance.

Source: NASA Black Marble User Guide, Collection 2.0 (2025)

The sum of lights, corrected step by step



No precautions

−70.7 M

fill −999.9 summed as if it were light

Final answer

22,737

national territory, filtered, water masked

Notebook 2 demo dataset (simulated values, Côte d'Ivoire, Collection 2 structure)

Mosaic, clipping and a skewed distribution

Mosaic

Same grid, same resolution: tiles joined by index, no distortion

Clipping

Point-in-polygon on the national border (official boundaries can replace the built-in ones)

Distribution

Log scale; median and sum tell different stories; check the extreme pixels

Area weighting

Pixel area ∝ cos(latitude): −13% at 30°, −20% at 37°

Exports GeoTIFF · CSV · NPZ · JSON metadata · processing log

SECTION

04

Analyse

How do we turn radiance into economic indicators?



One pipeline, six analysis scenarios

02 Parameters → **03** Acquire and clean → **04** Explore → **05–10** 6 scenarios → **11** Key messages → **12–13** Export and publish

S1 · Temporal dynamics

Is activity growing, and how fast?

Seasonally adjusted series · year-on-year · trend %/yr · anomalies · breaks

S2 · Regional growth

Which regions are growing, and are they converging?

Growth class · top 1 / top 10% shares · HHI · β and σ convergence

S3 · Lights and GDP

Do lights track GDP, and can they fill the gaps?

β elasticity · forecast · MAE vs naive · regional GDP

S4 · Shocks

How deep is the shock, how fast the recovery?

Trough % · deviation from trend · recovery month · placebo · daily index

S5 · Inequality and access

How unequal is light, and who lives in the dark?

Gini · Lorenz · Theil T · urban/rural ratio · population in the dark

S6 · Industrial zones

Where does the country produce?

Industrial score · zone share and trend · precision / recall vs OSM

Light data: four products, four roles

VIIRS VNL V2 · monthly

`NOAA/VIIRS/DNB/MONTHLY_V1/VCMSLCFG`

2014 → · ~464 m · avg_rad (nW·cm⁻²·sr⁻¹) + cf_cvg

Monthly radiance corrected for stray light; cf_cvg = clear nights

Role: S1 seasonality, S3 forecast, S4 shocks

VIIRS VNL V2.1 / V2.2 · annual

`NOAA/VIIRS/DNB/ANNUAL_V21 · _V22`

2013–2021 / 2022 → · ~464 m · average_masked

Annual radiance, background and ephemeral lights masked by EOG

Role: Annual levels for S1–S3, S5, S6; background mask

Black Marble VNP46A2 · daily

`NASA/VIIRS/002/VNP46A2`

2012 → · ~500 m · BRDF / Moon correction

Radiance for every night; quality flag 0–1, clear sky, snow-free

Role: S4 daily index around events

DMSP-OLS · harmonised

`NOAA/DMSP-OLS · sat-io Harmonized_NTL`

1992–2013 (+ DMSP-like VIIRS) · ~1 km · DN 0–63

Digital numbers, saturated at 63 in city centres; DN > 7 kept

Role: S1 long series, S3 long-run elasticity

Sources: Elvidge et al. (2017, 2021); Román et al. (2018); Li et al. (2020) · IDs checked against the Earth Engine catalogue on every run

Who lives there, and what has been built?

GHSL GHS-POP R2023A
`JRC/GHSL/P2023A/GHS_POP`
100 m · 1975–2030, 5-year epochs · population_count
Light per capita (S2, S5), inhabited pixels ≥ 5 people, population in the dark, flare test

WorldPop
`WorldPop/GP/100m/pop`
~100 m · 2000–2020 · population
Alternative population source (pop_source = "worldpop")

GHSL GHS-SMOD R2023A
`JRC/GHSL/P2023A/GHS_SMOD_V2-0`
1 km · 5-year epochs · smod_code (degree of urbanisation)
Urban if code ≥ 21 → urban / rural split of light and of population in the dark (S5)

GHSL GHS-BUILT-S R2023A
`JRC/GHSL/P2023A/GHS_BUILT_S`
100 m (from 10 m) · built_surface, built_surface_nres
Non-residential share of the industrial score (S6); "not built-up" flare test

ESA WorldCover v200
`ESA/WorldCover/v200`
10 m · 2021 · 11 land-cover classes
Optional: tree, crop, built-up, bare soil, water shares per unit — not used by default

Sources: Schiavina et al. (2023); Pesaresi et al. (2023), JRC GHSL R2023A; WorldPop (2018); Zanaga et al. (2022)

Boundaries, official statistics and validation data

Sentinel-5P TROPOMI NO_2

`COPERNICUS/S5P/OFFL/L3_NO2`

2018-07 → · ~1.1 km grid · tropospheric column (mol/m²)

Role: Annual mean per unit; Spearman ρ with light (S2); 20% of the industrial score (S6)

geoBoundaries CGAZ · FAO GAUL

`WM/geoLab/geoBoundaries/600` · `FAO/GAUL/2025`

ADM0 countries · ADM1 regions · ADM2 districts

Role: Units over which every pixel is summed; GAUL 2025 then 2015 if geoBoundaries fails

World Bank WDI (API)

`NY.GDP.MKTP.KD` · `EG.ELC.ACCS.ZS` · `SP.POP.TOTL` …

8 indicators · annual · national

Role: Real GDP (S3), electricity access total / urban / rural (S5), population check (±30%)

OpenStreetMap (Overpass)

`landuse = industrial`

Contributed polygons · completeness varies by country

Role: Validation only: precision and recall of candidate zones (S6)

Sources: Copernicus Sentinel-5P (ESA); Runfola et al. (2020); FAO GAUL 2025; World Bank WDI; © OpenStreetMap contributors (ODbL)

Which data feeds which scenario?

	Cleaning	S1	S2	S3	S4	S5	S6
VIIRS VNL monthly	●	●		●	●		
VIIRS VNL annual	●	●	●	●		●	●
Black Marble VNP46A2					●		
Harmonised DMSP	○	●		●			
GHS-POP / WorldPop	●		●			●	●
GHS-SMOD						●	
GHS-BUILT-S	●						●
Sentinel-5P NO_2			○				●
geoBoundaries / GAUL	●	●	●	●	●	●	●
World Bank WDI	○			●		○	
OpenStreetMap							○

● metric input ○ check, context or validation

From billions of pixels to two tables

Cleaned pixels

~464 m

↓

Sum per unit

ADM0 · ADM1 · ADM2

↓

Cached tables

+ parameter signature

Monthly table · unit × month

`sol`
sum of lights (observed + imputed)

`valid_frac`
share of the unit's pixels observed

`sol_adj`
sol if valid_frac ≥ 0.8, otherwise missing

`imputed_share`
share of light imputed (clouds)

Annual table · unit × year

`sol` · `sol_raw`
cleaned light · before flare mask

`lit_km2` · `area_km2`
area ≥ 0.5 nW · total area

`pop` · `pop_inhabited`
population · in pixels ≥ 5 people

`pop_dark_<t>`
inhabited, radiance < threshold t

`pop_urban` · `sol_urban`
population and light where SMOD ≥ 21

`flare_sol`
light removed as gas flaring

Remove everything that is not human activity

1 · Clouds

cf_cvg ≥ 2 · cloudy pixel = its annual value · unit-month kept if ≥ 80% observed

2 · Negatives

Set to 0: calibration noise over dark areas

3 · Outliers

Cap at 1,000 nW · > 5 × annual median → median (fires, lightning, glint)

4 · Background

Unlit in the annual product → 0; newly lit pixels kept if the year is not yet published

5 · Gas flares

≥ 50 nW, < 5 inhabitants, not built-up → removed with a 2 km halo

6 · DMSP

Saturation (DN 63) measured; harmonised series thresholded at DN > 7

Source: Li et al. (2020) for the DMSP threshold · all thresholds are NTLConfig parameters

Base metrics and checks, used everywhere

METRIC	DEFINITION / FORMULA	HOW TO READ IT
Sum of lights (SoL)	Σ radiance of the unit's cleaned pixels	Activity index; compare over time, with care across latitudes
Light per capita	SoL / population	Active area vs merely populated
Lit area share	area with radiance ≥ 0.5 nW / unit area	Spatial extent of electrification and settlement
Gas-flare share	flare_sol / sol_raw	High → oil-related light dominates; results less reliable
Imputed share	imputed light (cloudy pixels) / SoL	High → the month relies on annual values

- GHSL / World Bank population within [0.7; 1.3]
- VNL V2.1 → V2.2 jump vs neighbouring years
- DMSP → VIIRS jump (2013 → 2014)

Dynamics: is activity growing?

QUESTION

Is light rising, falling or flat — and is a recent move real or seasonal?

DATA USED

VIIRS VNL monthly
sol_adj per unit and per month, 2014 →

VIIRS VNL annual
sol per unit and per year

Harmonised DMSP–VIIRS
1992 →, to be read as a trend

METHOD, STEP BY STEP

1. **Regularise**
   Complete monthly calendar; gaps ≤ 3 months filled log-linearly
2. **Decompose (robust STL)**
   log(SoL) = trend + season (period 12) + remainder
3. **Seasonally adjust**
   SA = exp(trend + remainder): the curve to read
4. **Compare year on year**
   Each month vs the same month a year earlier
5. **Estimate the annual trend**
   Theil–Sen + Mann–Kendall on annual composites; regional series (ADM1)

Dynamics: the metrics

METRIC	DEFINITION / FORMULA	HOW TO READ IT
Seasonally adjusted SoL	exp(trend + remainder)	The level net of the usual seasonal profile
Year-on-year change	$SoL_m / SoL_{m-12} - 1$	The season cancels out; noisy from one month to the next
Annual growth (%/yr)	exp(Theil–Sen slope of log SoL) − 1, 95% CI	Robust trend; read the interval, not just the point
Mann–Kendall p-value	Test for a monotonic trend	Confidence: high $p < 0.01$ · medium $p < 0.10$ · low otherwise or $n < 6$
Anomaly	$\lvert r - \text{med}(r)\rvert / (1.4826 \cdot \text{MAD}) \geq 3.5$ and $\lvert\text{deviation}\rvert \geq 5\%$	An abnormal month, to be explained
Structural break	Binary segmentation, linear segments ≥ 12 months, BIC-type penalty	Lasting change in level or slope

Sources: Cleveland et al. (1990) STL; Sen (1968); Mann (1945), Kendall (1975)

Regions: who is growing, and are they converging?

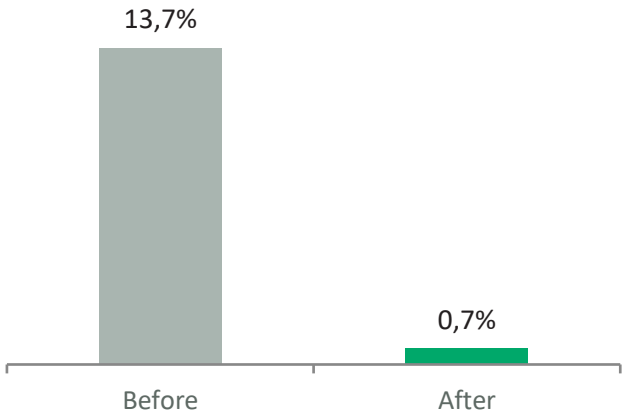
DATA USED

- Annual VNL SoL by ADM1 / ADM2
- GHS-POP → light per capita
- Sentinel-5P NO_2, latest year

METHOD, STEP BY STEP

1. **Robust growth per unit**
   Theil–Sen slope of log SoL, Mann–Kendall p
2. **Correct for multiple testing**
   Benjamini–Hochberg adjusted p (false discovery rate)
3. **Classify**
   Growth / Decline if |slope| ≥ 1%/yr and adjusted p < 0.10, otherwise Stagnation
4. **Concentration and convergence**
   Top shares, HHI, β (3-year initial mean), σ; NO_2 correlation

Spurious "convergence" under pure noise



Simulation: 60 units, 12 years, 25% noise, no real convergence — share of draws falsely significant at 95%

Regional growth: the metrics

METRIC	DEFINITION / FORMULA	HOW TO READ IT
Growth class	$\|g\| \geq 1\%/yr$ and BH-adjusted $p < 0.10$	Growth / Decline / Stagnation (= not established)
Top 1 and top 10% shares	SoL of the brightest unit(s) / total SoL	Weight of the capital or main hub
Herfindahl index (HHI)	Σs_i^2 $(s_i = SoL_i / \Sigma SoL)$	0 → dispersed · 1 → all light in one unit
β-convergence	$g_i = a + \beta \cdot \log(\text{initial light per capita}_i) + \varepsilon$, HC1 errors	$\beta < 0$ with CI < 0 → lagging units are catching up
Half-life	$\ln 2 / -\ln(1 + \beta)$	Years to close half of the gap
σ-convergence	standard deviation of log(light per capita) across units, per year	Falling σ → regions are becoming more alike
NO_2 correlation	Spearman ρ (mean NO_2, SoL) across units	High ρ → light reflects combustion activity

Sources: Benjamini & Hochberg (1995); Barro & Sala-i-Martin (1992); Herfindahl (1950)

Lights and GDP: measuring the elasticity

QUESTION

Does light follow official GDP, and by how much does it move when GDP moves by 1%?

DATA USED

VIIRS VNL annual, national
SoL per year (+ V2.1 / V2.2 dummy)

Harmonised long series
1992 →, sensor dummy

World Bank NY.GDP.MKTP.KD
Real GDP, constant 2015 USD

METHOD, STEP BY STEP

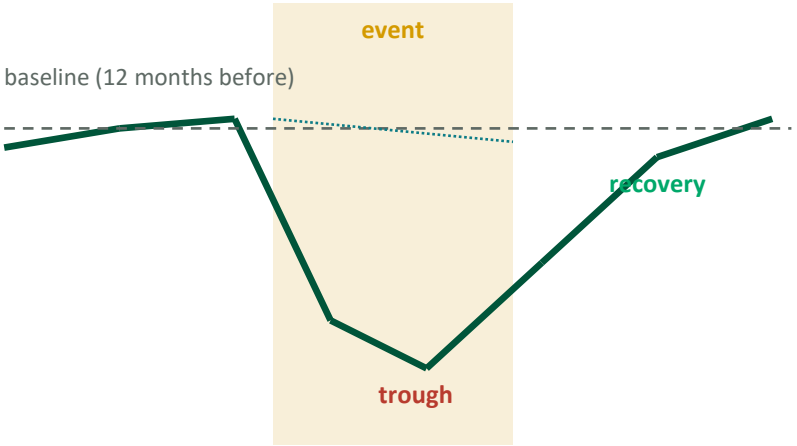
1. **Levels**
   $\log(SoL_t) = \alpha + \beta \cdot \log(GDP_t) [+ \delta \cdot V22_t] + \varepsilon_t$
2. **Growth rates**
   $\Delta\log(SoL) = \alpha + \beta \cdot \Delta\log(GDP)$: immune to common trends
3. **Long series**
   Levels from 1992 → with a DMSP / VIIRS dummy
4. **Uncertainty**
   95% Newey–West CI (HAC) + paired bootstrap CI (999 draws)
5. **Diagnostics**
   Durbin–Watson · Breusch–Pagan · Jarque–Bera · Ramsey RESET ($n \geq 12$)

Lights and GDP: forecasting and regional estimates

METRIC	DEFINITION / FORMULA	HOW TO READ IT
β elasticity	slope of log SoL on log GDP (or growth rates)	+1% GDP ⟷ +β% light; Henderson's 0.3 ≈ 1/β
Confidence in β	high: n ≥ 15, p < 0.01, CI width < 1, R^2 ≥ 0.6	medium if p < 0.10 · low if n < 8 or not significant
Nowcast	Δlog GDP = a + γ·Δlog SoL, chained from the last official year	Estimates for unpublished years, 80 / 95% bands
Pseudo real-time MAE	mean |actual − predicted| growth, in points; expanding window ≥ 5 years	Typical error of a forecaster using lights
Relative MAE	MAE lights / min(MAE last year, MAE past mean)	< 1 → lights add more than naive rules
Regional GDP (experimental)	$GDP_i = GDP \cdot SoL_i^{\gamma} / \Sigma SoL_j^{\gamma}$, γ = 1; 0.5	Sum = national GDP; γ=1 vs 0.5 gap = uncertainty

Sources: Henderson, Storeygard & Weil (2012); Newey & West (1987) · regional GDP is labelled experimental everywhere

Shocks: how deep, how long, and is it real?



Schematic: seasonally adjusted series around an event

DATA USED

Seasonally adjusted monthly VNL (S1) · daily Black Marble VNP46A2 · catalogue: COVID-19, CUSTOM_EVENTS, ACLED / EM-DAT

METRIC	FORMULA	READING
Trough %	100·(exp(min SA − baseline) − 1)	Depth of the fall
Deviation from trend %	mean deviation from the trend of the previous 24 months	Loss vs the expected path
Recovery month	first month SA ≥ baseline (≤ 36)	Speed of recovery; none = not recovered
Placebo test	trough < 5th percentile of earlier windows	Stands out from ordinary noise

Shocks: daily zoom and regional impact

Keep the good nights

Quality flag ∈ {0, 1} · cloud bits 6–7 ≤ 1 · snow-free · pixel lit in the annual product

Daily sum

Day kept if ≥ 50% of pixels observed; SoL scaled to the observed share

Smooth

7-day rolling median: absorbs lunar residuals and cloudy days

Index

100 × SoL / same days 364 days earlier (same weekday)

Reading the daily index

100	same activity as a year earlier
< 100	activity below last year (outage, lockdown, conflict)
> 100	activity above last year (recovery, electrification)

Regional impact and unexplained breaks

- Heat map: trough % by ADM1 × event (−30% → +30%).
- Breaks with no catalogued event within ±6 months → document before interpreting.

Inequality: is light unevenly distributed?

QUESTION

Is light per capita concentrated in a few places, and at what level should we act?

DATA USED

VIIRS VNL annual
SoL by district (finest level)

GHS-POP
population weights, light per capita

GHS-SMOD
urban if code ≥ 21

METHOD, STEP BY STEP

1. **Light per capita by district**
   $x_i = SoL_i / pop_i$, weights $w_i = pop_i$
2. **Gini and Lorenz curve**
   Cumulative population share vs cumulative light share
3. **Theil T and its decomposition**
   T = between regions (ADM1) + within regions
4. **Change and its uncertainty**
   ΔGini first → last year, 300 paired bootstrap draws, 95% CI
5. **Urban / rural gap**
   Urban / rural light per capita

Population in the dark and electricity access

METRIC	DEFINITION / FORMULA	HOW TO READ IT
Gini	weighted; 0 = equality, 1 = all light in one unit	Overall concentration of light per capita
Theil T	$\Sigma\, w_i\, (x_i/\mu)\, \ln(x_i/\mu)$, w_i population shares	Decomposes into between- and within-region parts
Urban / rural ratio	light per capita where SMOD ≥ 21 ÷ elsewhere	How much more light an urban resident gets
Population in the dark	Σ pop of pixels ≥ 5 people with radiance < t	Published for t = 0.25 · 0.5 · 1 nW: a range
Share in the dark	pop in the dark / inhabited population	Proxy for the share without electricity
Gap to surveys (pts)	(1 − share in the dark) − WDI access rate EG.ELC.ACCS.ZS	Large gap → low-consumption access or a data problem

Sources: Theil (1967); Schiavina et al. (2023) GHSL; World Bank WDI · gas-flare halos excluded from the count

Industrial zones: four signals, one score per pixel

SIGNAL	DATA	NORMALISATION (0–1)	WEIGHT
Non-residential built-up	GHSL BUILT-S	nres share / 0.6	**35 %**
Light intensity	VNL, median of last 3 yrs	$\log_{10}$(radiance) / 2	**30 %**
NO_2 emissions	Sentinel-5P, annual mean	(NO_2 − med) / (p98 − med)	**20 %**
Few residents	GHSL GHS-POP	1 − $\log_{10}(1 + pop/km^2) / \log_{10}(5000)$	**15 %**



Industrial score

$$S = 0.35 \cdot built_nres + 0.30 \cdot light + 0.20 \cdot NO_2 + 0.15 \cdot few_residents \quad \text{(each signal bounded to 0–1)}$$

Weights and thresholds: expert choices in IND.PARAMS, to be tested · Sources: Pesaresi et al. (2023); Copernicus Sentinel-5P

Industrial zones: from pixels to validated candidates

METRIC	DEFINITION / FORMULA	HOW TO READ IT
Candidate pixel	radiance ≥ 5 nW, built-up ≥ 1%, S ≥ 0.55 — or gas flare	Neighbours (≥ 2 pixels) → zone; 250 zones max
Share of national light	Zone SoL / national SoL (latest year)	Economic weight of the zone
Zone dynamics	Theil–Sen growth: > +5%/yr expansion, < −3% decline	Signal of opening, expansion or slowdown
Class	energy: flare share > 0.3 and nres < 0.3 · major: top 10% SoL or ≥ 8 km²	Energy site · major hub · industrial / logistics zone
Precision	zones overlapping an OSM industrial area / zones	≥ 0.5 → medium confidence, otherwise low
Recall	OSM industrial areas found / OSM areas	Coverage; depends on OSM completeness
Weighted precision	SoL of confirmed zones / SoL of all zones	Is most of the industrial light confirmed?

From metrics to key messages

High — Established result: quotable with its interval

Medium — Plausible result: quote with caution and context

Low — A lead, not a conclusion: verify before use

Messages are generated in EN and FR and fill the notebook's executive summary (section 00)

WHERE CONFIDENCE COMES FROM

Trend (S1, S2)

high $p < 0.01$ · medium $p < 0.10$ · low if $n < 6$

Elasticity (S3)

n, p, CI width, R^2 (see S3)

Shock (S4)

placebo test passed or not

Inequality (S5)

the ΔGini bootstrap CI excludes 0

Industry (S6)

OSM precision ≥ 0.5 → medium, otherwise low

SECTION

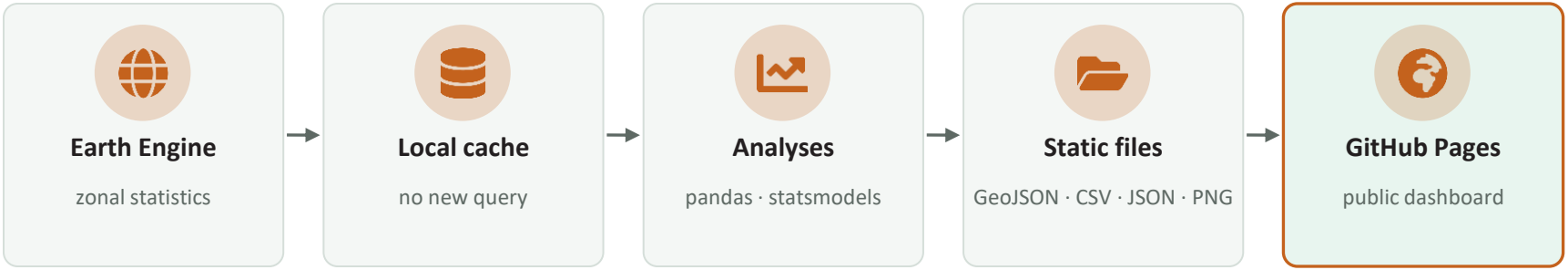
05

Publish and check

How do results reach users, and how far can they be trusted?



A static, bilingual dashboard, in one call



Bilingual EN / FR

Same numbers, text generated in each language

Reproducibility manifest

Parameters, versions, data availability

Secure publishing

Limited-scope token, 7–30 days, written nowhere

Key messages with confidence: high · medium · low

Synthetic test country "XTS": public publishing refused

Verified before release

0

runtime errors, notebooks 1 and 2, EN and FR

13.7%→0.7%

spurious β-convergence under pure noise (simulation)

29 → 0

spurious growth classes across 300 null districts

- Attributes and fill values read from every file
- Parameters and code version invalidate a stale cache
- Every export travels with its metadata and log
- Quality legend and water codes read, not assumed
- GHSL population checked against the World Bank (±30%)
- Simulated values always flagged as such

Six artefacts to know before drawing conclusions

Blooming

Light spills beyond the built-up area

Saturation

The brightest centres hit a ceiling; DMSP saturates at 63

Gas flares

Bright with no local economic activity

Seasonality

Clouds and fires create non-economic cycles

Sensor breaks

DMSP → VIIRS, VNL V2.1 → V2.2, collections

Rural noise

Weakest signal where the rural poor live

Sources: Gibson et al. (2021); Elvidge et al. (2021); Li et al. (2020)

SECTION

06

Key takeaways

What should a statistician remember?



Twelve checks before publishing a number

1. File names intact; year read from the name
2. Dataset group searched for, not assumed
3. Scale, offset and fill read from the file
4. Fill set to NaN first
5. North up: row 0 is the northernmost
6. Quality threshold documented
7. Oceans and seas masked
8. Number of nights examined
9. Brightest pixel is plausible
10. Same layer and version across the whole series
11. The right product for the question
12. Limits written down

Trend on A4, season on A3, event on A2



Choose the product

Trend → annual A4 · season → monthly A3 · dated event → daily A2. Never the other way round.



Respect the order

Fill → NaN, quality, water, mosaic, clipping. Every skipped step silently changes the number.



Complement, don't replace

Direction, timing and geography of activity — always alongside official statistics and documented limits.

Sources and references

- Román, M. O. et al. (2018). NASA's Black Marble nighttime lights product suite. Remote Sensing of Environment 210, 113–143.
- NASA (2025). Black Marble User Guide, Collection 2.0. LAADS DAAC / Earthdata.
- Elvidge, C. D. et al. (2017). VIIRS night-time lights. Int. J. Remote Sensing 38(21).
- Elvidge, C. D. et al. (2021). Annual time series of global VIIRS nighttime lights. Remote Sensing 13(5).
- Li, X. et al. (2020). A harmonized global nighttime light dataset 1992–2018. Scientific Data 7.
- Henderson, J. V., Storeygard, A., Weil, D. N. (2012). Measuring economic growth from outer space. AER 102(2).
- Chen, X., Nordhaus, W. D. (2011). Using luminosity data as a proxy for economic statistics. PNAS 108(21).
- Gibson, J. et al. (2021). Which night lights data should we use in economics, and where? JDE 149.
- Cleveland, R. B. et al. (1990). STL: a seasonal-trend decomposition procedure based on Loess. JOS 6(1).
- Schiavina, M. et al. (2023). GHS-POP R2023A; Pesaresi, M. et al. (2023). GHS-BUILT-S R2023A. JRC.
- Runfola, D. et al. (2020). geoBoundaries. PLoS ONE 15(4) · Natural Earth (public domain).
- World Bank, World Development Indicators · ESA WorldCover 2021 · Copernicus Sentinel-5P · OpenStreetMap contributors.

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AfDB × DATA INNOVATION FOR AFRICA (DI4A)

One ISO3 code in, a dashboard out

Collect · Explore and understand · Analyse and publish



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